

Computational biology

Lecture II: Building genomes

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Statistiques pour les sciences du Vivant et de l'Homme

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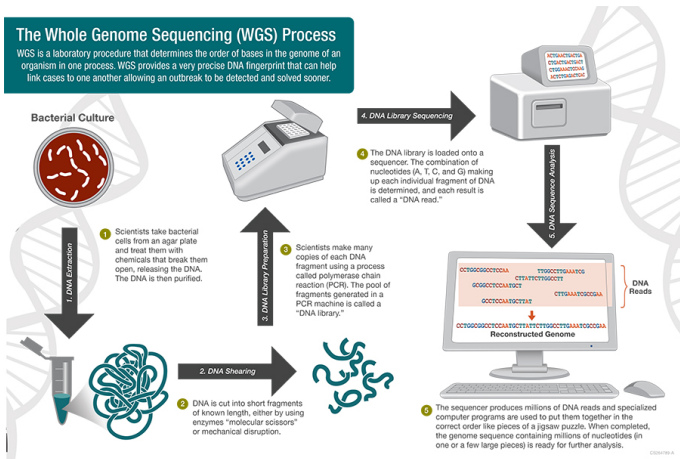
Various types of sequencing for various scientific questions

- Whole genome sequencing
 - Discovery of gene responsible for hereditary disease
- Metagenomics
 - Discovery of genomes and biological function of a microbial sample
- Metabarcoding/Amplicon
 - Discovery/monitoring of presence of organisms in a sample

Sequencing steps WGS

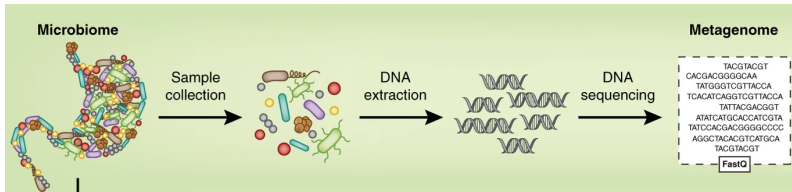
The Whole Genome Sequencing (WGS) Process

WGS is a laboratory procedure that determines the order of bases in the genome of an organism in one process. WGS provides a very precise DNA fingerprint that can help link cases to one another allowing an outbreak to be detected and solved sooner.



Source cdc.gov

Metagenomics: best suited to study bacteria



[Quince et al. Nat. Biotech. 17]

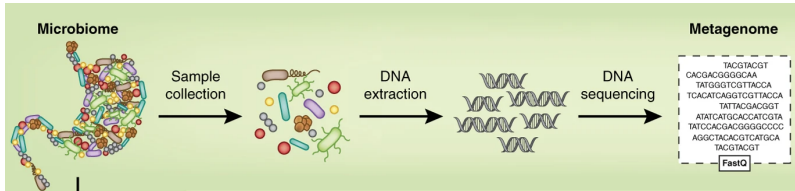
Almost the same process for sequencing, but with many (microbial) organisms in the sample.

The downstream analysis will differ.

Scope

For most environmental samples, metagenomics data will consists mostly in prokaryotic data, with a little bit of viral and eukaryotic.

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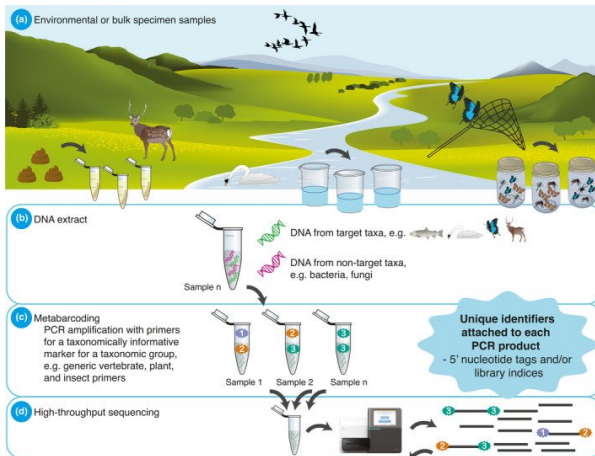
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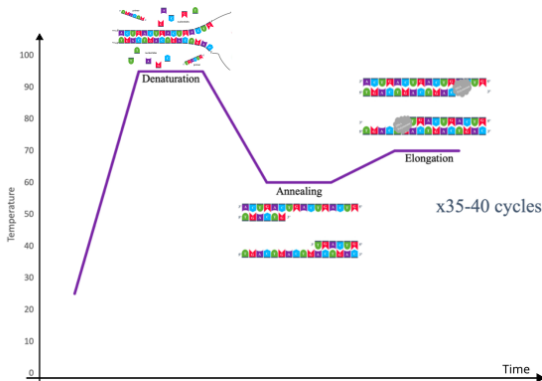
Not suitable when interested for instance of monitoring e.g. insects in an ecosystem.

Sequencing steps amplicon for targeting specific taxa



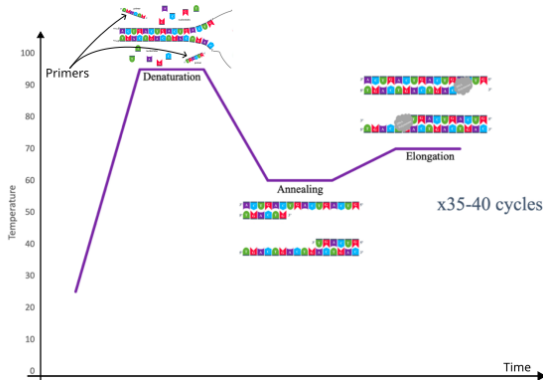
Adapted from [Bohmann et al. Mol. Ec. Res. 21]

Polymerase Chain Reaction: amplify DNA



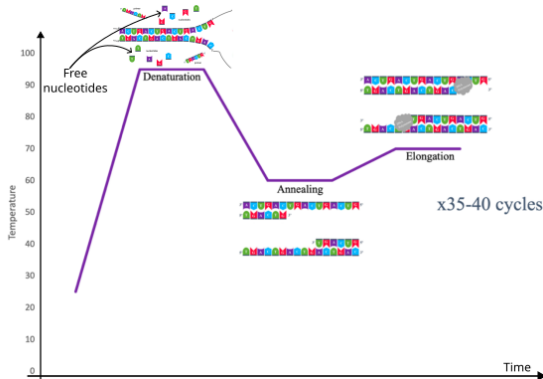
Crucial Elongation step carried by *Thermus aquaticus* polymerase (Taq).
Adapted from here

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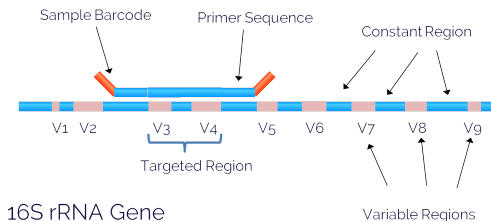


Crucial Elongation step carried by *Thermus aquaticus* polymerase (Taq).
Adapted from here

The choice of primers

Primers are either:

- Synthesized and specific to a branch of the tree of life (amplicon)



- Random primers that will amplify everything

Sequencing technologies

Beyond the sequencing technique (WGS/amplicon), there are now three main sequencing technologies:

Technology	Method	Read length
Illumina	Optical	short
PacBio	Optical	long
Minlon	Nanopore	long

DNA is very tiny: how to read it?

Visible wavelength:

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Visible wavelength: 380nm (violet)-750nm (red)

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We need indirect ways of "reading" the bases.

Optical reading: very low intensity, high noise

To "read" the DNA optically, the idea is to synthesize complementary DNA strand using fluorescent-modified nucleotides, with one specific color for each base.

Optical reading: very low intensity, high noise

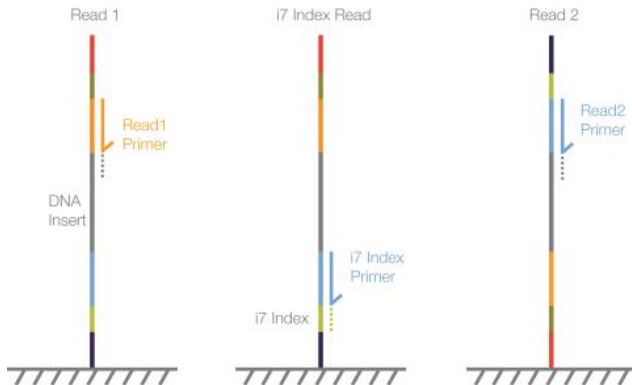
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Issue: Very low signal, high noise

- Illumina: increase the signal by copying the same DNA fragment many times locally
- PacBio: reduce the noise using a ZMW optical well

Illumina 2010'S

<https://www.youtube.com/watch?v=EDVKxSNdSic>

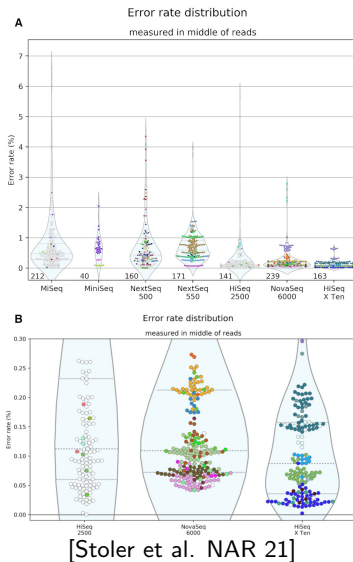


Main features of Illumina data

As a first order approximation:

- Reads of length $2 \times 150\text{bp}$ - $2 \times 300\text{bp}$
- Almost all errors are substitution of nucleotides
- Errors are uniformly distributed along the genome, but more probable at the end of the read
- Reads are uniformly distributed along the genome

Very low error rate for Illumina



Errors are (slightly) biased toward the previous base



[Stoler et al. NAR 21]

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Longer homopolymers

Illumina reports sequencing quality

Together with the nucleic sequence, Illumina reports sequencing quality based on the optical intensity and signal-to-noise ratio (doc). This optical quality is translated into a quality score Q (Phred score) related to the probability P of error:

$$Q = -10 \cdot \log_{10}(P)$$

Does it work?

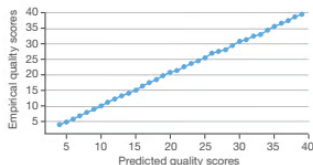
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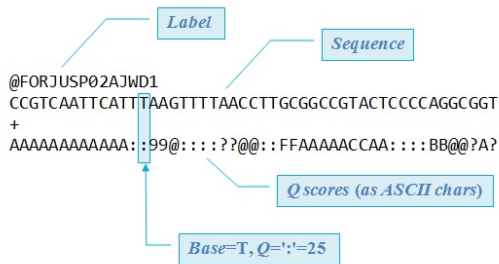
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Figure 1: High Correlation of Empirical and Predicted Q Scores

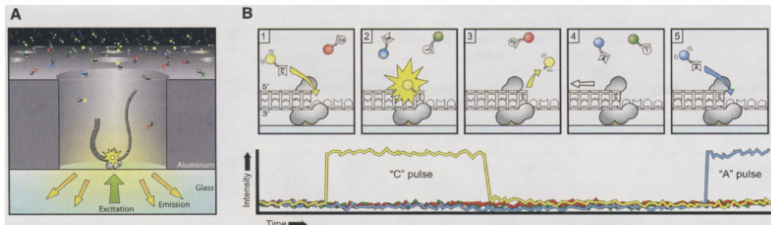


Illumina sequencing Q scores are highly accurate. This example shows that predicted Q scores for a HiSeq 2000 run correlate well to empirically derived Q scores.

Sequence and quality stored in FASTQ files



Long read technology



ZMW video HiFi PacBio sequencing video

Nanopore



images/nanopore.gif

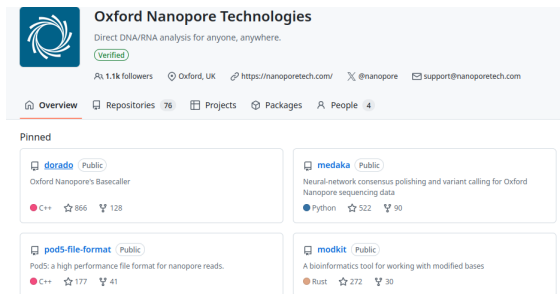
images/nanopore_signal.gif

Much improvement came from base calling

Base calling is the process of identifying from the raw (electrical) signal the currently sequenced nucleotide.

Nanopore develops open-source Dorado software that employs a machine learning model (using LibTorch) to detect nucleotides from raw signal.

Also provides open-source software to train custom models (link)



The screenshot shows the GitHub profile of Oxford Nanopore Technologies. The profile header includes the company logo, name, a tagline, a 'Verified' badge, and statistics for followers, repositories, projects, packages, and people. Below the header, the 'Pinned' section displays four repositories: **dorado** (Basecaller, C++, 866 stars, 128 forks), **medaka** (Polishing, Python, 522 stars, 90 forks), **pod5-file-format** (File format, C++, 177 stars, 41 forks), and **modkit** (Bioinformatics tool, Rust, 272 stars, 30 forks).

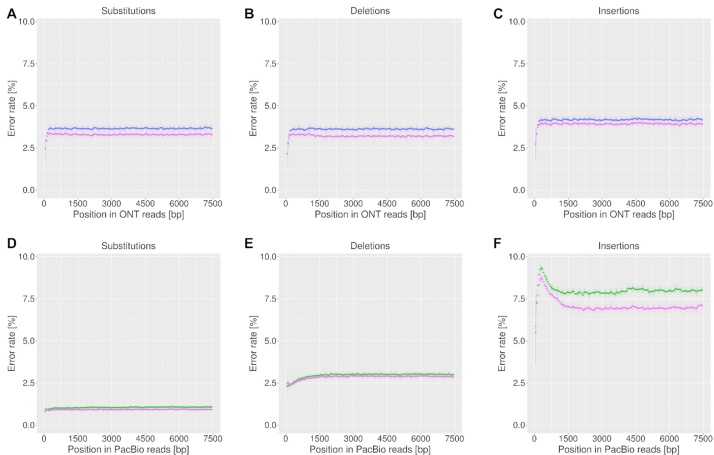
Oxford Nanopore Technologies
Direct DNA/RNA analysis for anyone, anywhere.
Verified
1.1k followers · Oxford, UK · <https://nanoporetech.com/> · @nanopore · support@nanoporetech.com

Overview · Repositories 76 · Projects · Packages · People 4

Pinned

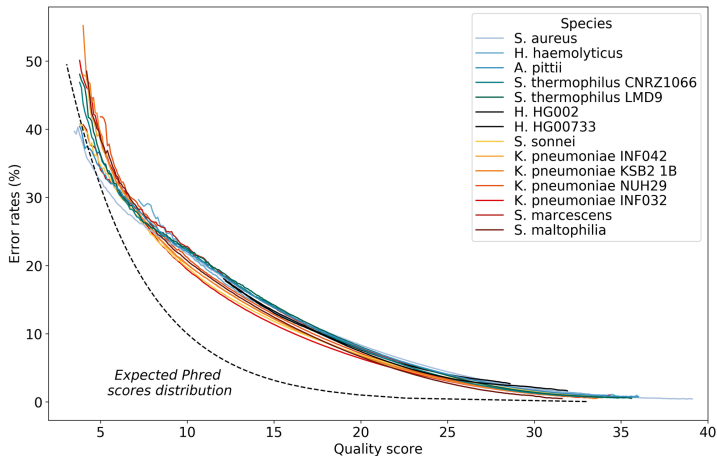
- dorado** (Public)
Oxford Nanopore's Basecaller
C++ · 866 stars · 128 forks
- medaka** (Public)
Neural-network consensus polishing and variant calling for Oxford Nanopore sequencing data
Python · 522 stars · 90 forks
- pod5-file-format** (Public)
Pod5: a high performance file format for nanopore reads.
C++ · 177 stars · 41 forks
- modkit** (Public)
A bioinformatics tool for working with modified bases
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More and more diverse errors in long-reads



[Dohm et al. NAR 20]

Phred quality badly calibrated but informative

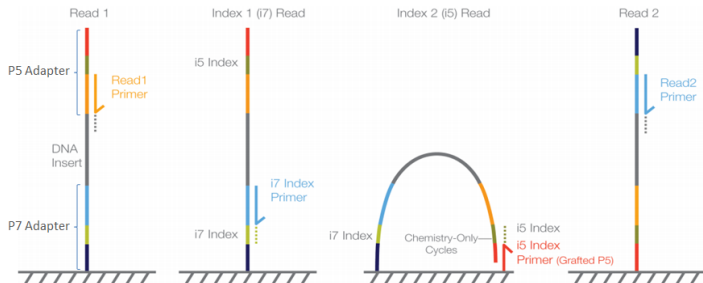


Delahaye & Nicolas, PLOS One 21

Bioinformatics processing of sequencing data

Read cleaning

- Reads need to be filter by quality (*FastQC*, *Fastp*)
- Trimming: Reads need to be clipped from adapter (*Trimmomatic*, *Cutadapt*)



[illumina.com]

Amplicon

(A lot) of noise in amplicon data

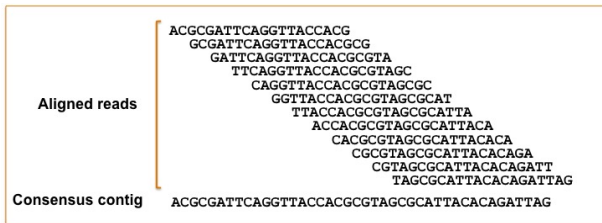
Beyond sequencing errors, amplicon relies on amplification, which is very much error-prone.

- Formation of chimeras
- Mutations during DNA copy
- Sensitive to contamination
- Still unknown artifact

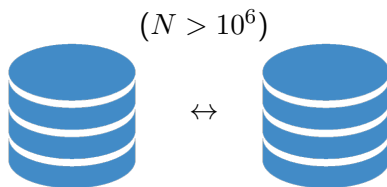
Several tools exist (dada2, vsearch, obitools) and are being developed to correct these errors.

Assembly - short reads

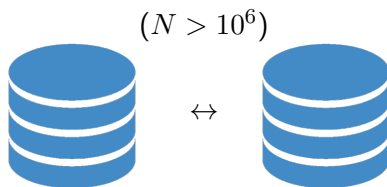
Basic principle: stack overlapping fragments



Naïve overlapping detection does not scale

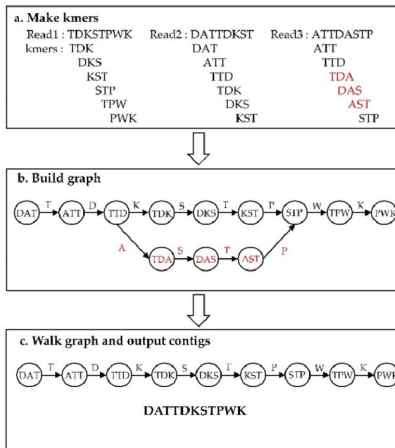


Naïve overlapping detection does not scale



Complexity?

DeBruijn graphs



[Lu et al 23]

DeBruijn graphs leads linear time

k-mer (nodes of the graph) can be hashed and accessed in linear time.
Choose heuristics such that the walk can be achieved in linear time.

Leads to a linear-time algorithm

Empirically $k=31$ is a good choice

If k is *too small* $\rightarrow$

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Need many heuristics to make an assembler

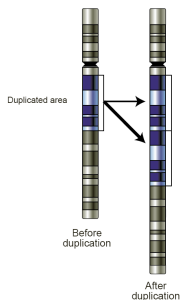
One needs to implement heuristics so that the walk in the graph output correct contigs

Plus, many implementation variants (speed, memory footprint, etc.). Few examples of ideas:

- You can store only the nodes, and not the edges
- Storing the nodes can be achieved using a Bloom Filter for low memory footprint (Minia assembler)
- Consider multiple DeBruijn graphs (with various k) (SPAdes assembler)

Small read issues: repeats and structural variants

Since short read sequencing provides only a local information about the genome sequence, larger scale features such as repeats or copy number variations may be missed or break the assembly.



[National Human Genome Research Institute]

Assembly - hybrid (long+short reads)

Top-down and bootom-up

Hybrid assemblies can be done either:

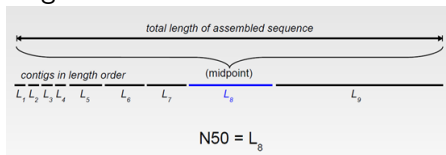
- Assembling short reads *de novo* and chaining them using long-read information
 - Best option if long-read data has low coverage
 - Software: Opera-MS, Unicycler (for bacterial genomes), SPAdes
- Assembling using long reads and polishing by mapping short reads.
 - Software: Long-read assembler (CANU,Flye) + Pilon/SNP caller

Assembly - benchmark

Metrics to assess assembly quality

Walk in the branching DeBruijn graph leads to fragmented assemblies. To assess synthetically the fragmentation of the genome assembly we commonly use the following metrics:

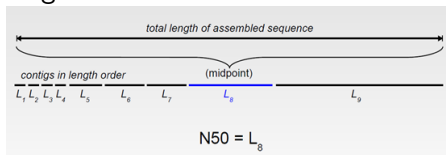
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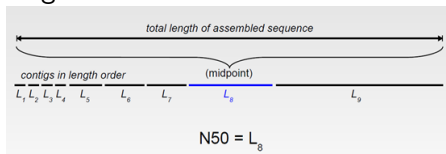


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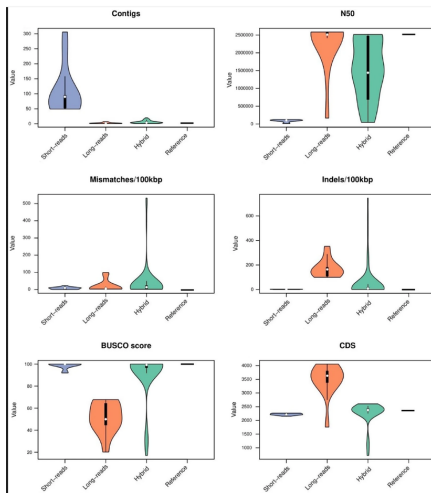
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- NG50: same idea but the total length is taken externally (from a reference genome).
- L50: minimal number of contigs to make 50% of the total assembled length

Hybrid strategy is the closest to reference



Assembly of *Xylella fastidiosa*, in violins: several seq. depth and several algorithms. [Rojas-Miranda et al. Sci. Rep. 25]

Specific case of metagenomics

When it comes to metagenomics, many software are not suitable.

See [Goussarov et al Microb. 24] for a benchmarking in a real complexity case.

